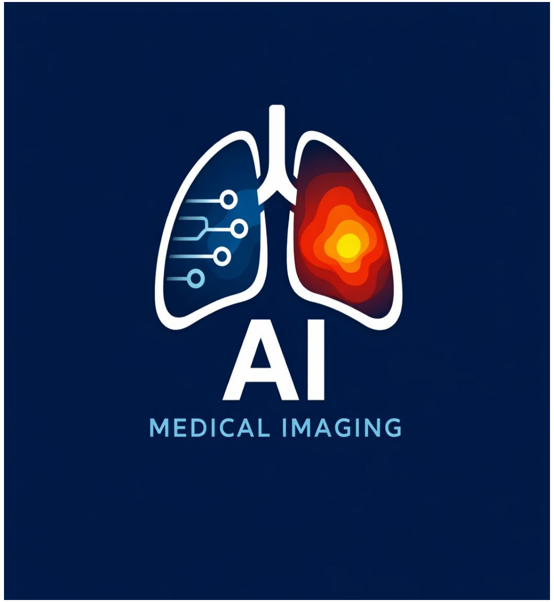


““Multi-Label Deep Learning for Differentiating Pneumonia and Pleural Effusion in Chest X-ray images ”.

Deep Learning Assisting Radiology

Presented by: TEFFAHI Manel &GACEMI Raida &MEFTAH Sara

2025

Problem

Millions of patients suffer from respiratory diseases such as pneumonia and pleural effusion, which require early diagnosis to avoid serious complications.

Chest X-rays are widely used to detect these diseases, but interpreting medical images can be challenging because it requires experienced radiologists and can be time-consuming.

According to the World Health Organization (WHO), pneumonia remains one of the leading causes of death worldwide, affecting millions of patients every year.

AI-based medical imaging tools can help support doctors by analyzing X-ray images automatically.

Target Customers

- Our solution targets healthcare institutions that analyze chest X-ray images on the daily basis .

- Primary customers:
  - Hospitals with radiology departments
  - Radiology clinics performing diagnostic imaging

- Future expansion:
  - We aim to deploy our system on telemedicine platforms that provide remote diagnostic services.

Team

- **Team :**
- Meftah Sara
- Gacemi Raida Aicha
- Teffahi Manel Zineb

Medical Advisor:.Radiology specialist (future collaboration)

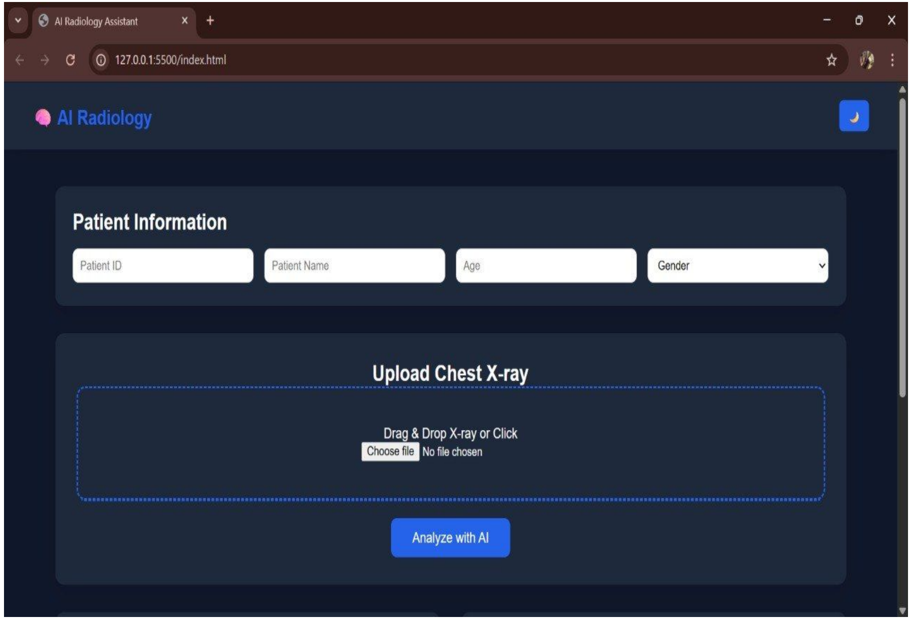
Our Innovation

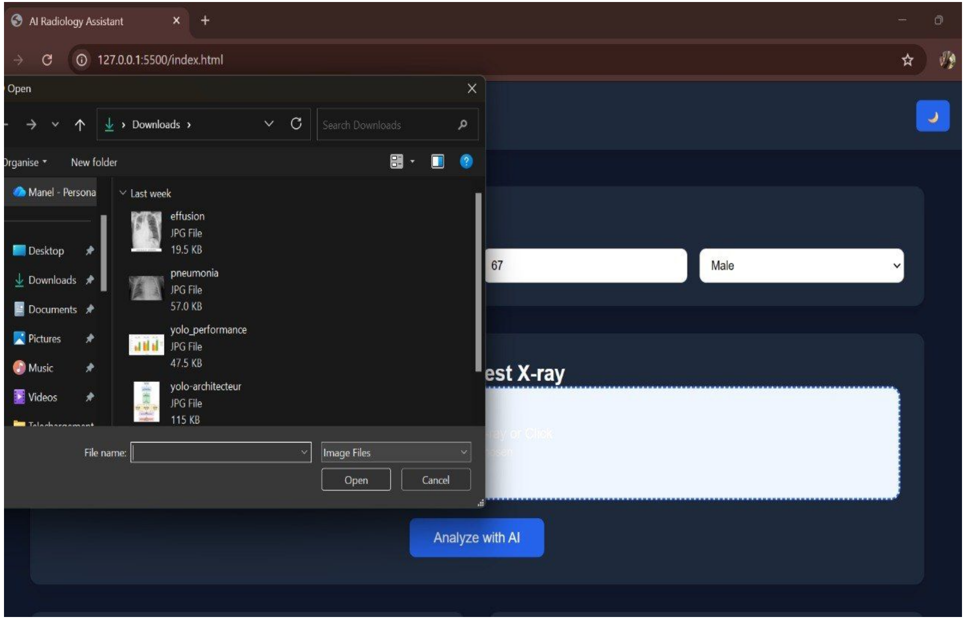
- We introduce an AI-powered assistant for chest X-ray analysis.
- The system can quickly detect possible lung diseases such as pneumonia and pleural effusion.
- It provides an instant preliminary analysis to support doctors.
- Our innovation helps reduce diagnosis time and workload for radiologists.
- It improves efficiency by combining medical expertise with artificial intelligence.
- The goal is not to replace doctors, but to enhance their decision-making process.

Our Innovation - How It Works (Continue)

- The system operates through the following steps:

1- A chest X-ray image is uploaded to the system.





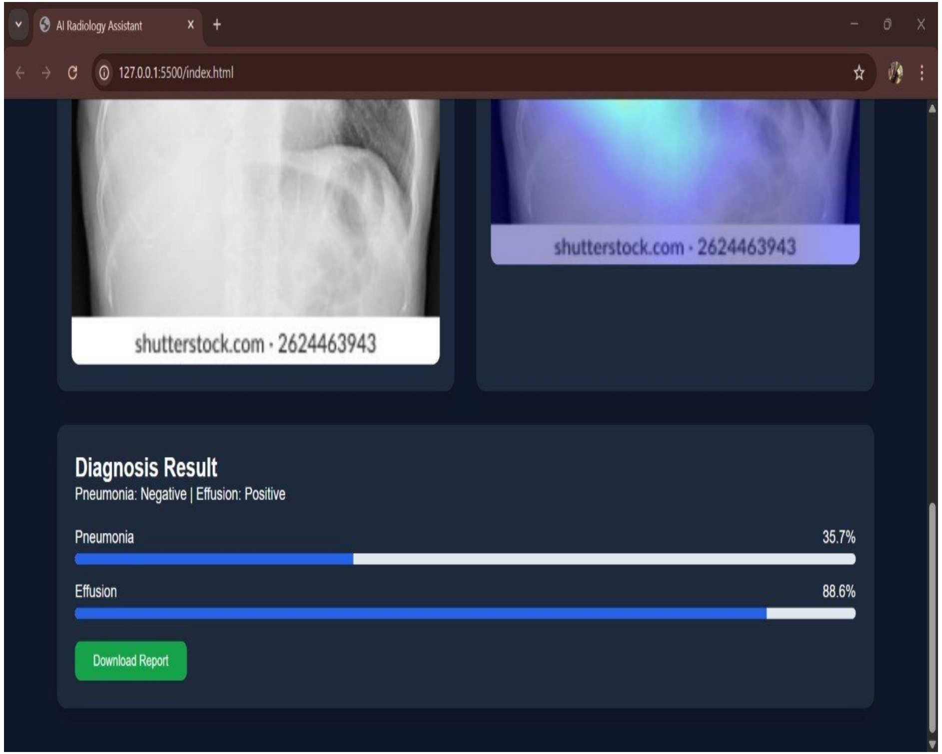
Our Innovation - How It Works (Continue)

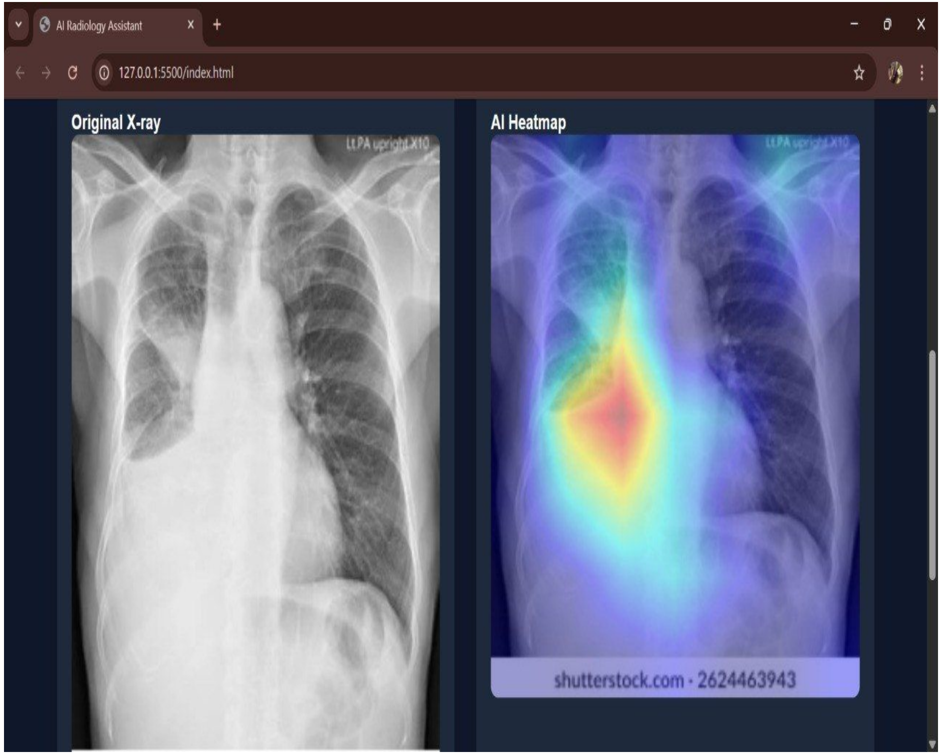
2- The image is preprocessed:resized and normlized

3- The image is analyzed by a deep convolutional neural network (DenseNet-121) trained on medical imaging datasets.

4-The model outputs predictions indicating the probability of: Pneumonia , Pleural Effusion.

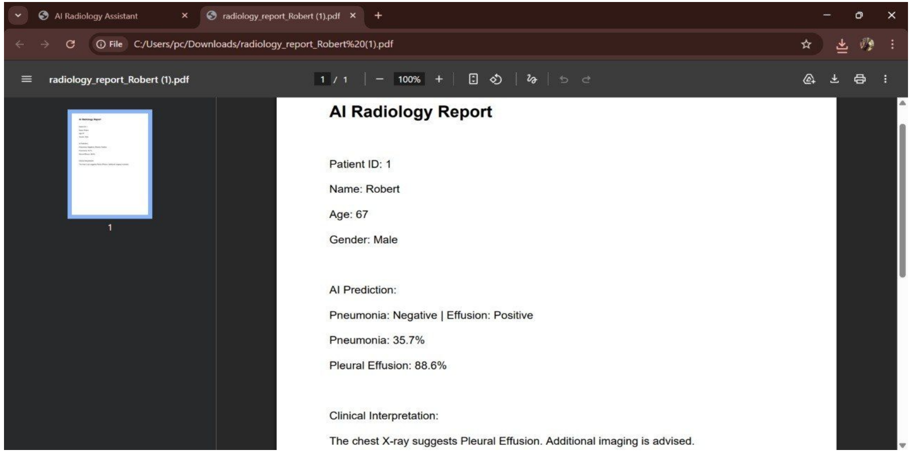
Our Innovation - How It Works (Continue)





Our Innovation - How It Works (Continue)

- Finally, the system provides automated diagnostic repport to assist radiologists.



- Based on the report, which should always be reviewed by a medical professional, the system helps doctors prioritize high-risk cases.

Our Innovation - Why It Works (Continue)

Our system is based on the hypothesis that CNN models can learn complex visual patterns directly from chest X-ray images. These models automatically extract hierarchical features, starting from simple edges and textures to high-level patterns such as **lung opacities (linked to pneumonia)** and **fluid accumulation (linked to pleural effusion)**, which may not always be easily noticeable to the human eye.

This approach is strongly supported by research in medical image analysis and deep learning, where CNNs have demonstrated high performance in detecting various diseases from imaging data.

Our Innovation - Why It's Better

Traditional Consultations	Proposed AI System
Chest X-rays must be manually analyzed by a radiologist.	automatically analyzes the X-ray image and detects patterns related to pneumonia and pleural effusion.
Diagnosis may take time and a large numbers of images can increase pressure on medical staff.	AI can process large volumes of medical images quickly and consistently., helping doctors prioritize urgent cases.
Limited number of radiologists in some regions.	AI assists doctors in areas with few specialists.
Patients may need to travel to another city for interpretation.	AI can support local radiology centers, reducing the need for travel.
Interpretation depends entirely on human workload and human expertise..	AI acts as a decision-support tool, and does not replace doctors.

Prototype

Our prototype was developed using Python and deep learning frameworks.

Key technologies used:

- PyTorch deep learning library
- Medical imaging dataset (CheXpert)
- GPU-based training environment

Prototype implementation is available through our kaggle development environment.

https://www.kaggle.com/code/manelteffahi/multi-label-dl-pneumonia-effusion-1
https://www.kaggle.com/code/sarameftah/multi-label-dl-pneumonia-effusion

Benchmark

To evaluate the performance of our model, we used standard machine learning evaluation metrics:

- AUROC (Area Under the ROC Curve)
- F1 Score

These metrics measure how well the model distinguishes between disease present and disease absent cases.

Benchmarking helps compare the effectiveness of different model configurations.

Results

- The trained model achieved the following validation performance:
- Pneumonia detection:
- AUROC ≈ 0.741
- Pleural Effusion detection:
- AUROC ≈ 0.926
- These results indicate that the model performs better at detecting pleural effusion than pneumonia, highlighting areas for future improvement.

Scalability Plan

- To scale the solution, several steps are planned:

- Testing the model on larger medical datasets
- Increasing training epochs for better performance
- Conducting validation with real hospital data

- Future deployment could integrate the system into hospital radiology workflows and medical imaging platforms.

Business Sides

Value Proposition

Our AI-powered diagnostic support system helps hospitals, clinics, and radiologists overcome delays and heavy workloads

Unlike traditional workflows, our solution delivers faster preliminary results, supports medical decision-making, improves efficiency, and enhances patient care while allowing healthcare professionals to focus on critical cases.

Market Analysis

Although AI-based systems for detecting pneumonia and pleural effusion from chest X-rays are increasingly available worldwide, their adoption remains concentrated in high-income healthcare systems. Most existing solutions are designed for hospitals with advanced digital infrastructure (PACS systems, cloud-based integration, and high-performance computing), which limits their usability in developing countries such as Algeria. In addition, these solutions are often expensive and subscription-based, making them inaccessible for public hospitals and unaffordable for individual patients. Furthermore, limitations in healthcare digitization and payment systems further restrict access to international AI platforms. As a result, there is a strong market gap for low-cost, locally deployable AI diagnostic tools adapted to resource-limited healthcare environments.

Market Analysis

Market Segment	2024 Value (USD)	Growth Trend (CAGR)	Source
AI in medical imaging in Algeria	early, nascent stages		Journal of Economic Growth and Entrepreneurship JEGE
Regional Context (Middle East & Africa)	US$ 179.9	CAGR of 32.8%	The Algerian and Comparative Public Law Journal Vol. 11, n° 01/ July 2025, pp 226-234
AI in global Chest Imaging Market	~USD 1.88 billion	nearly 37%.	https://www.fortunebusinessinsights.com/ai-in-medical-imaging-market-

Cost in Algeria

Type	Chest X-ray Cost	Interpretation Time	Specialist Availability	Total Impact
Public Hospital (with Chifa card)	Free	Several hours or more waiting	Limited radiologists; patients may travel to another city	Long waiting time for diagnosis
Private Radiology Clinic	1500-2000 DZD	Around 1 hour or more	Radiologist available but workload may cause delays	Faster but still requires waiting
Proposed AI-Assisted System	Same X-ray cost 500dzd for each right diagnosis	AI analysis in a few seconds	Supports radiologists with automatic pre-analysis	Faster diagnosis and better case prioritization

Competitive Landscape

Current solutions include:

- Traditional radiology interpretation by doctors
- Commercial AI platforms for medical imaging (e.g., AI radiology tools)

Limitations of current systems:

- High cost of commercial AI solutions
- Limited availability in developing countries
- Lack of specialized models for local healthcare systems

Go-To-Market Strategy

Our deployment strategy focuses on local radiology centers and public healthcare institutions in Algeria, where there is often a shortage of radiologists and pneumologists.

In cities such as Saïda, patients sometimes need to travel to larger cities to have their chest X-rays interpreted due to limited specialist availability.

Go-To-Market Strategy

Initial Target Locations

Private radiology centers such as:

- **Basso Imaging Center**

These centers often operate with a limited number of radiologists, creating delays in image interpretation.

Our AI system could assist by providing preliminary analysis of chest X-rays, helping radiologists prioritize urgent cases, **while also enabling centers to handle significantly more patients per day (e.g., increasing from ~30 to 60), with a shared-value model based on the added revenue.**

Go-To-Market Strategy

Public Healthcare Opportunity(future work)

Public hospitals in Algeria also face challenges:

- Limited number of radiologists and pneumologists
- High number of patients needing imaging exams
- Delays in diagnosis due to specialist shortages

Our system could be used as a decision-support tool in public hospitals, helping doctors quickly identify suspected pneumonia or pleural effusion cases.

Business Model

Our solution follows a B2B model (Business-to-Business) targeting radiology centers and hospitals in Algeria.

Revenue Options

1- Pilot Deployment Contracts (per-scan model)

Radiology centers can start with a pilot program where the AI system is tested in their workflow to analyze chest X-ray images.

They pay a small fee per analyzed X-ray image.

This model allows small clinics to adopt the technology without high upfront costs.

Business Model

2-Hospital Collaboration Projects

Public hospitals could adopt the system through government healthcare innovation programs and university partnerships. In the future, we aim for governmental support to provide stronger computing resources, enabling training on larger datasets with more epochs to improve performance. After deployment, the model would be regularly updated and improved (mise à jour) to maintain accuracy. The system could also be expanded into a broader platform for detecting multiple diseases.

Partnership

Medical Experts

Radiologists and pneumologists will help:

- Validate model predictions
- Provide medical feedback
- Improve system reliability

Team

Name / Nom	Role / Rôle	Expertise & Background / Expertise et parcours
Meftah Sara	**Project Manager**	Coordination, documentation
Raida Gacemi	**Data Analyst / ML Support**	Data preprocessing, model support
Teffahi Manel	**ML Engineer**	Python, model training, evaluation
Dr. Abderrahmane Khiat	**Founder, R&D**	PhD in Computer Science – expert in, AI systems, and digital health. Former Project Lead at Fraunhofer IAIS and IT Project Manager at Hyundai AutoEver.

Contact

Dr. Abderrahmane Khiat
Founder & Project Lead

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